An-Najah National University
Master of AI
IoT Principles & Design
Course Project- Phase#1
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Deadline: 10/11/2025

Objective:
This assignment will help you understand the basic principles of IoT systems through simulation. You will model a simple *Smart Home* environment with **sensors**, a **central computer (controller)**, and **actuators**.

Scenario:
Imagine a smart home equipped with various IoT devices:

- **Sensors**: Devices that sense the environment (e.g., temperature, light, motion, humidity, gas).
- **Central Computer (Controller)**: Collects sensor data, analyzes it, and makes decisions (On the cloud).
- **Actuators**: Devices that execute commands (e.g., turn on/off lights, fans, alarms, or sprinklers).

The simulation should generate **random sensor data** (within realistic ranges) and demonstrate how the central computer makes **decisions** based on this data, then sends corresponding **commands** to actuators.

Requirements:

1. **Sensors**
    - Simulate at least **3 types of sensors** (e.g., temperature, motion, light).
    - Each sensor should periodically generate random values.
    - Example ranges:
        - Temperature: 15–40 °C
        - Light: 0–100 (0 = dark, 100 = bright)
        - Motion: 0 (no motion) or 1 (motion detected)
2. **Central Computer (Controller)**
    - Collects all incoming sensor readings and store them in a data warehouse on the cloud.
    - Implements **decision-making rules** (e.g., *If temperature > 30, turn on fan*; *If motion detected and light < 20, turn on light*).
    - Outputs log messages to the console (e.g., “Temperature = 35 → Fan ON”).
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3. **Actuators**
   - Simulate at least **2 actuators** (e.g., fan, light, alarm).
   - The central computer controls them based on sensor input.
   - Display actuator status (ON/OFF).
4. **Simulation**
   - Run for a fixed period (e.g., 20 cycles).
   - In each cycle: sensors send data → central computer analyzes → actuators respond.
5. **Implementation Notes:**
   - You may use **Python, Java, or C++** for implementation.
   - Make your code modular (separate classes for sensors, controller, actuators).
   - Clearly print sensor readings, decisions, and actuator states at each step.

Deliverables:

1. Source code with clear documentation.
2. A short report (2–3 pages) including:
   - Description of your design (sensor types, rules, actuator roles).
   - Example output logs.
   - Reflection: how this models a real IoT system.